

Class 8: Cancer Mini Project

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Background

In today's class we will apply the methods and techniques clustering and PCA to help make sense of a real world breast cancer FNA biopsy data set.

Data Import

We start by importing our data. It is a CSV file so we will use the `read.csv()` function.

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names =1)
```

have a wee peak at the first few rows"

```
head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1
84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1

	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean
842302	0.11840	0.27760	0.3001	0.14710
842517	0.08474	0.07864	0.0869	0.07017
84300903	0.10960	0.15990	0.1974	0.12790
84348301	0.14250	0.28390	0.2414	0.10520
84358402	0.10030	0.13280	0.1980	0.10430
843786	0.12780	0.17000	0.1578	0.08089

	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
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842302	0.2419		0.07871	1.0950	0.9053	8.589
842517	0.1812		0.05667	0.5435	0.7339	3.398
84300903	0.2069		0.05999	0.7456	0.7869	4.585
84348301	0.2597		0.09744	0.4956	1.1560	3.445
84358402	0.1809		0.05883	0.7572	0.7813	5.438
843786	0.2087		0.07613	0.3345	0.8902	2.217
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se	
842302	153.40	0.006399	0.04904	0.05373		0.01587
842517	74.08	0.005225	0.01308	0.01860		0.01340
84300903	94.03	0.006150	0.04006	0.03832		0.02058
84348301	27.23	0.009110	0.07458	0.05661		0.01867
84358402	94.44	0.011490	0.02461	0.05688		0.01885
843786	27.19	0.007510	0.03345	0.03672		0.01137
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst		
842302	0.03003		0.006193	25.38		17.33
842517	0.01389		0.003532	24.99		23.41
84300903	0.02250		0.004571	23.57		25.53
84348301	0.05963		0.009208	14.91		26.50
84358402	0.01756		0.005115	22.54		16.67
843786	0.02165		0.005082	15.47		23.75
	perimeter_worst	area_worst	smoothness_worst	compactness_worst		
842302		184.60	2019.0	0.1622		0.6656
842517		158.80	1956.0	0.1238		0.1866
84300903		152.50	1709.0	0.1444		0.4245
84348301		98.87	567.7	0.2098		0.8663
84358402		152.20	1575.0	0.1374		0.2050
843786		103.40	741.6	0.1791		0.5249
	concavity_worst	concave.points_worst	symmetry_worst			
842302		0.7119	0.2654	0.4601		
842517		0.2416	0.1860	0.2750		
84300903		0.4504	0.2430	0.3613		
84348301		0.6869	0.2575	0.6638		
84358402		0.4000	0.1625	0.2364		
843786		0.5355	0.1741	0.3985		
	fractal_dimension_worst					
842302		0.11890				
842517		0.08902				
84300903		0.08758				
84348301		0.17300				
84358402		0.07678				
843786		0.12440				

Make sure to remove the first `diagnosis` column - I don't want to use this for my machine

learning models. We will use it later to compare our results to the expert diagnosis.

```
wisc.data <- wisc.df[,-1]
diagnosis <- wisc.df$diagnosis
```

Exploratory Data Analysis

Q1. How many observations are in this dataset?

```
nrow(wisc.data)
```

```
[1] 569
```

Q2. How many of the observations have a malignant diagnosis?

```
sum(wisc.df$diagnosis == "M")
```

```
[1] 212
```

```
table(wisc.df$diagnosis)
```

```
  B   M
357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#colnames(wisc.data)
length(grep("_mean$", colnames(wisc.data)))
```

```
[1] 10
```

Principal Component Analysis

The main functions here is `prcomp()` and we want to make sure we set the optional argument `scale=TRUE` :

```
wisc.pr <- prcomp(wisc.data, scale=T)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

PC2 = 0.1897, there are getting smaller and smaller.

If you add up all the PCs..., the cumulative of 2 PCs is 0.6324 (63%)

Q4. From your results, what proportion of the original variance is captured by the first principal component (PC1)?

PC1 captures 0.4427, 44%

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

3 PCs is 72%

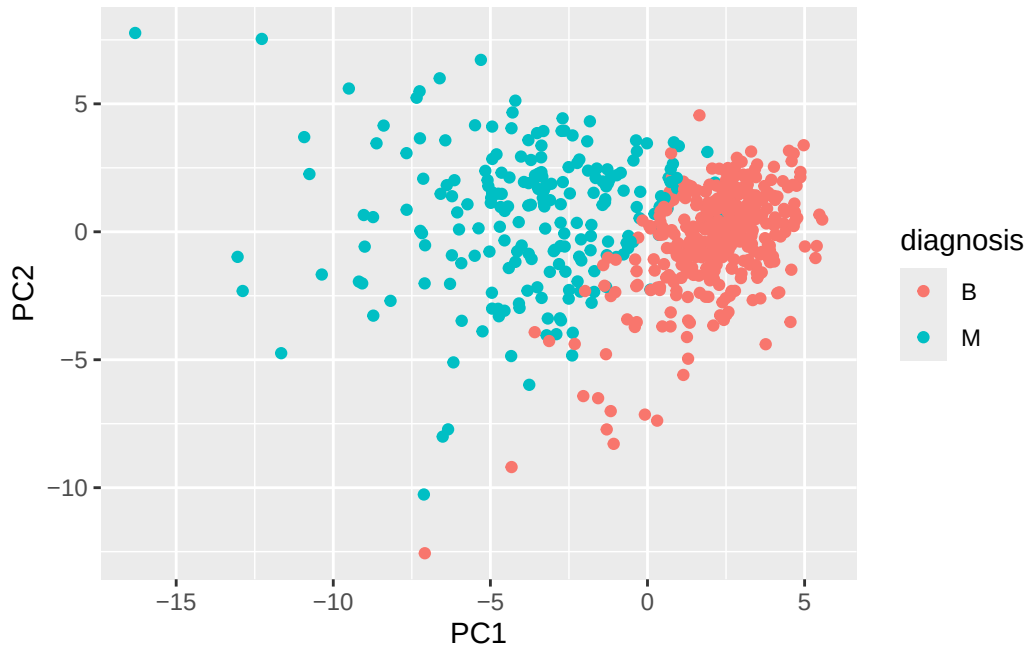
Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

7 PCs is 91%

Our main PCA “score plot” or “PC plot” of results:

```
library(ggplot2)
```

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC2, col=diagnosis) +  
  geom_point()
```



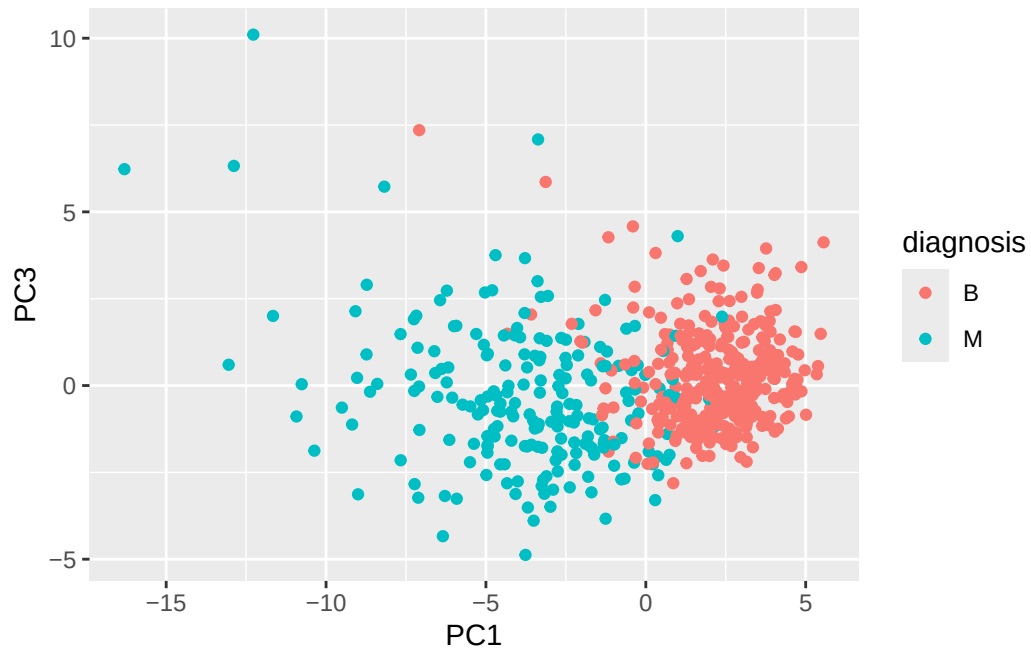
Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

Each point is a person, M = malignant B = Benign

- similar characteristics will cluster together - Cell PC1 score (cell1 # influence on PC1 #)+()
- low weight PCs don't contribute/influence the cells much compared to the higher weighted PCs

Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots? - I notice that they are more towards the bottom of the graph.

```
ggplot(wisc.pr$x) +  
  aes(PC1, PC3, col=diagnosis) +  
  geom_point()
```



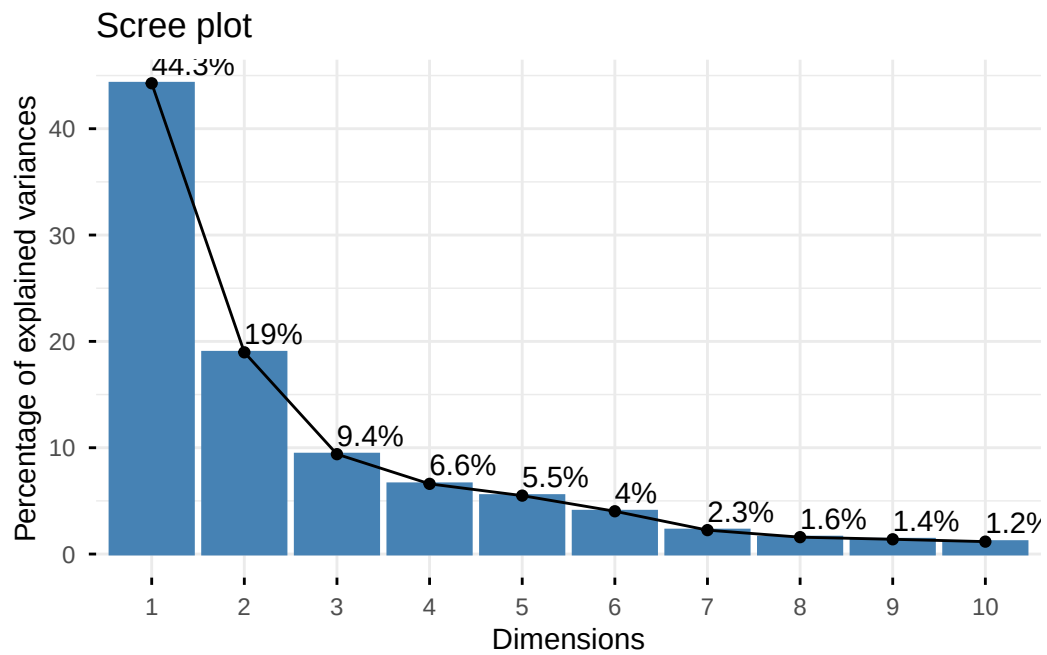
Scree-plot:

```
library(factoextra)
```

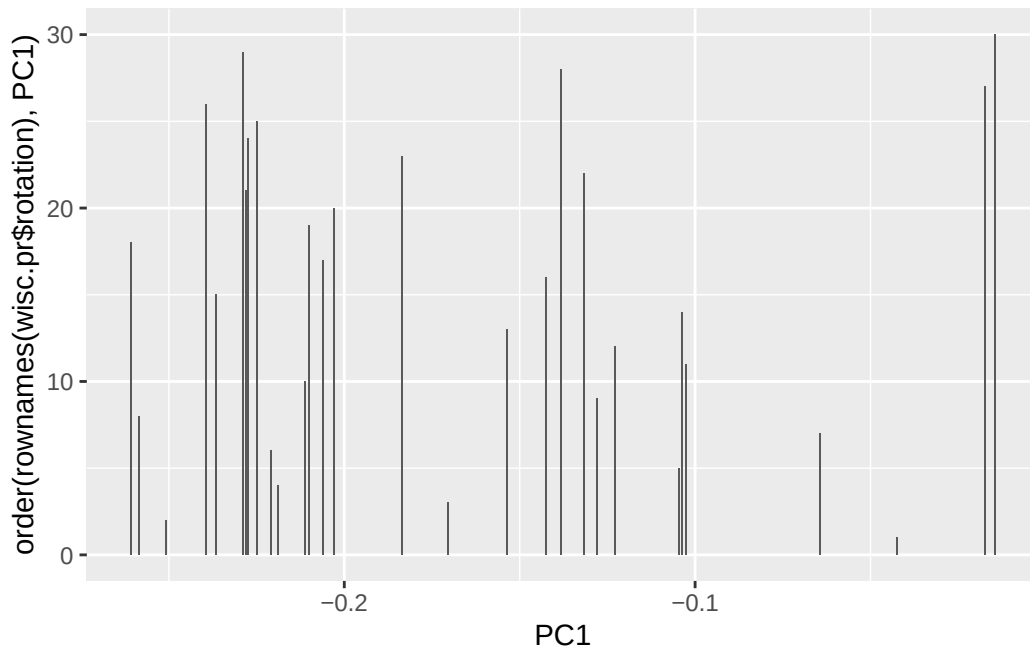
Welcome! Want to learn more? See two factoextra-related books at <https://goo.gl/ve3WBa>

```
fviz_eig(wisc.pr, addlabels= TRUE)
```

Warning in geom_bar(stat = "identity", fill = barfill, color = barcolor, :
Ignoring empty aesthetic: `width`.



```
ggplot(wisc.pr$rotation)+  
  aes(PC1,  
       order( rownames(wisc.pr$rotation), PC1)) +  
  geom_col()
```



Collectively these two plot (“score plot” and “loading plot”) tells us that if cells nucleus are deeply indented (“conclave”), irregular circular (“compactness”), and have large “perimeter” values they tend to be malignant...

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`? This tells us how much this original feature contributes to the first PC. Are there any features with larger contributions than this one?

```
wisc.pr$rotation["concave.points_mean",1]
```

```
[1] -0.2608538
```

Hierarchical Clustering

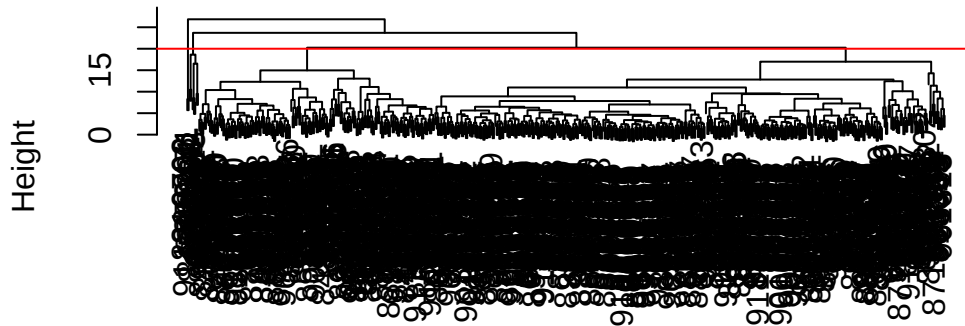
First scale the data (with the `scale()` function), then calculate a distance matrix (with the `dist()` function). Then cluster with the `hclust()` function and plot”

```
wisc.hclust <- hclust(dist(scale(wisc.data)))
```

Q10. Using the `plot()` and `abline()` functions, what is the height at which the clustering model has 4 clusters?

```
plot(wisc.hclust)
abline(h=20, col="red")
```

Cluster Dendrogram



```
dist(scale(wisc.data))
hclust(*, "complete")
```

You can also use the `cutree()` function with a argument `k=4` rather than `h=height`

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters)
```

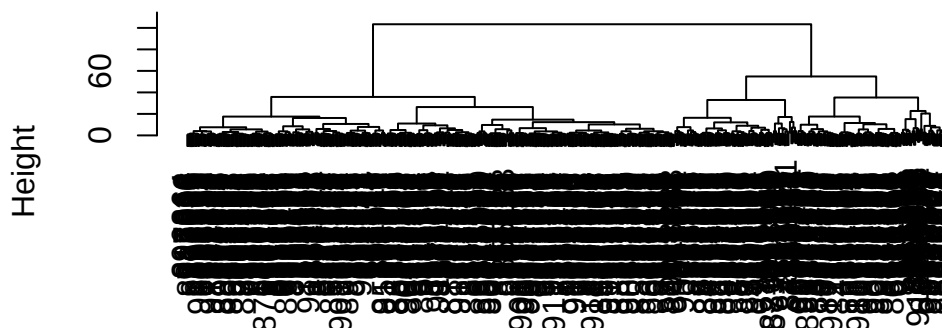
```
wisc.hclust.clusters
  1  2  3  4
177  7 383  2
```

Combining methods

Here we will take our PCA results and use those input for clustering. In other words our `wisc.pr$x` scores that we plotted above (the main output from PCA - how the data lie on our new principal component axis/variables) and use a subset of the PCs that capture the most variance as input for `hclust()`

```
pc.dist <- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(pc.dist, method="ward.D2")
plot(wisc.pr.hclust)
```

Cluster Dendrogram



```
pc.dist
hclust (*, "ward.D2")
```

Cut the dendrogram/tree into two main group1/clusters:

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

```
grps
  1   2
203 366
```

I want to know how the clustering in `grps` with values of 1 or 2 correspond the expert diagnosis.

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1   24 179
  2  333   33
```

My clustering **groups 1** are mostly “M” diagnosis (179) and my clustering **group 2** are mostly “B” diagnosis

24 FP 179 TP 333 TN 33 FN

Sensitivity $TP/(TP+FN)$

```
(179/(179+33))
```

```
[1] 0.8443396
```

Sensitivity $TN/(TN+FP)$

```
333/(333+24)
```

```
[1] 0.9327731
```

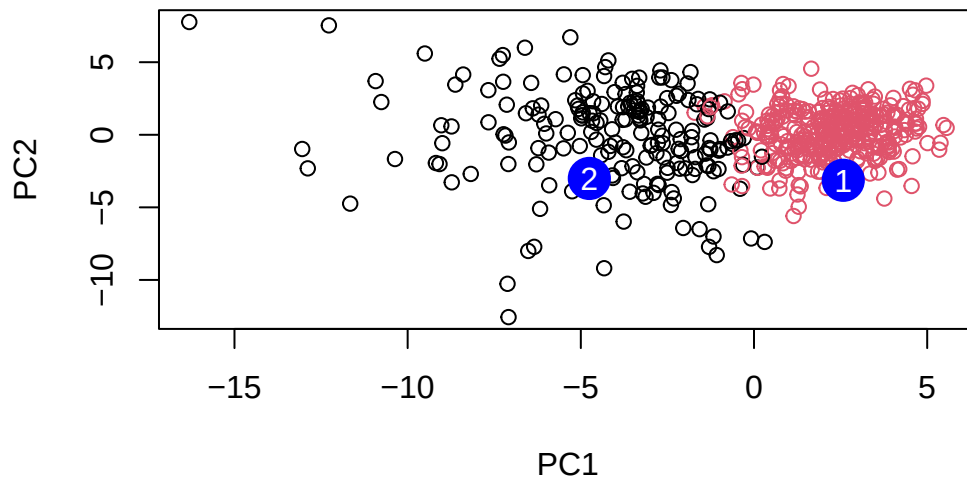
Q15. OPTIONAL: Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

There is a high specificity and strong sensitivity compared to clustering on the original data.

```
#url <- "new_samples.csv"
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
plot(wisc.pr$x[,1:2], col=grps)
points(npc[,1], npc[,2], col="blue", pch=16, cex=3)
text(npc[,1], npc[,2], c(1,2), col="white")
```



Q16. Which of these new patients should we prioritize for follow up based on your results? - We should prioritize the patient with cluster 1 for a follow up because they are malignant.